Assignment-8

Tushar Sharma

Regression Diagnostics and Hat Matrix Analysis

Objective

1. Understand the mathematical structure of linear regression.
2. Compute the **Hat Matrix** manually.
3. Identify **leverage and influential observations**.
4. Evaluate regression assumptions using diagnostic plots and statistical tests

Dataset

- - - - Boston Housing Data set - - - -

Target variable:

```
MEDV → Median value of owner-occupied homes
```

Predictor examples:

```
RM, LSTAT, CRIM, NOX
```

- Loading libraries

```r
library(dplyr)
```

```
Attaching package: 'dplyr'
```

```
The following objects are masked from 'package:stats':

    filter, lag

The following objects are masked from 'package:base':

    intersect, setdiff, setequal, union
```

```r
library(ggplot2)
library(car)
```

```
Loading required package: carData


Attaching package: 'car'

The following object is masked from 'package:dplyr':

    recode
```

```r
library(lmtest)
```

```
Loading required package: zoo


Attaching package: 'zoo'

The following objects are masked from 'package:base':

    as.Date, as.Date.numeric
```

```r
library(nortest)
library(qpcR)
```

```
Loading required package: MASS


Attaching package: 'MASS'
```

```
The following object is masked from 'package:dplyr':

    select

Loading required package: minpack.lm

Loading required package: rgl

Loading required package: robustbase

Loading required package: Matrix
```

```r
setwd("D:/Sem-2/R-/7,8,9,10/7")
data<-read.csv("HousingData.csv")
head(data)
```

```
     CRIM ZN INDUS CHAS   NOX    RM  AGE    DIS RAD TAX PTRATIO      B LSTAT
1 0.00632 18  2.31    0 0.538 6.575 65.2 4.0900   1 296    15.3 396.90  4.98
2 0.02731  0  7.07    0 0.469 6.421 78.9 4.9671   2 242    17.8 396.90  9.14
3 0.02729  0  7.07    0 0.469 7.185 61.1 4.9671   2 242    17.8 392.83  4.03
4 0.03237  0  2.18    0 0.458 6.998 45.8 6.0622   3 222    18.7 394.63  2.94
5 0.06905  0  2.18    0 0.458 7.147 54.2 6.0622   3 222    18.7 396.90    NA
6 0.02985  0  2.18    0 0.458 6.430 58.7 6.0622   3 222    18.7 394.12  5.21
  MEDV
1 24.0
2 21.6
3 34.7
4 33.4
5 36.2
6 28.7
```

```r
str(data)
```

```
'data.frame':   506 obs. of  14 variables:
 $ CRIM   : num  0.00632 0.02731 0.02729 0.03237 0.06905 ...
 $ ZN     : num  18 0 0 0 0 0 12.5 12.5 12.5 12.5 ...
 $ INDUS  : num  2.31 7.07 7.07 2.18 2.18 2.18 7.87 7.87 7.87 7.87 ...
 $ CHAS   : int  0 0 0 0 0 0 NA 0 0 NA ...
 $ NOX    : num  0.538 0.469 0.469 0.458 0.458 0.458 0.524 0.524 0.524 0.524 ...
 $ RM     : num  6.58 6.42 7.18 7 7.15 ...
```

```
$ AGE    : num  65.2 78.9 61.1 45.8 54.2 58.7 66.6 96.1 100 85.9 ...
$ DIS    : num  4.09 4.97 4.97 6.06 6.06 ...
$ RAD    : int  1 2 2 3 3 3 5 5 5 5 ...
$ TAX    : int  296 242 242 222 222 222 311 311 311 311 ...
$ PTRATIO: num  15.3 17.8 17.8 18.7 18.7 18.7 15.2 15.2 15.2 15.2 ...
$ B      : num  397 397 393 395 397 ...
$ LSTAT  : num  4.98 9.14 4.03 2.94 NA ...
$ MEDV   : num  24 21.6 34.7 33.4 36.2 28.7 22.9 27.1 16.5 18.9 ...
```

```r
summary(data)
```

```
     CRIM                ZN             INDUS            CHAS        
 Min.   : 0.00632   Min.   :  0.00   Min.   : 0.46   Min.   :0.00000  
 1st Qu.: 0.08190   1st Qu.:  0.00   1st Qu.: 5.19   1st Qu.:0.00000  
 Median : 0.25372   Median :  0.00   Median : 9.69   Median :0.00000  
 Mean   : 3.61187   Mean   : 11.21   Mean   :11.08   Mean   :0.06996  
 3rd Qu.: 3.56026   3rd Qu.: 12.50   3rd Qu.:18.10   3rd Qu.:0.00000  
 Max.   :88.97620   Max.   :100.00   Max.   :27.74   Max.   :1.00000  
 NA's   :20         NA's   :20       NA's   :20      NA's   :20       
     NOX               RM             AGE              DIS        
 Min.   :0.3850   Min.   :3.561   Min.   :  2.90   Min.   : 1.130  
 1st Qu.:0.4490   1st Qu.:5.886   1st Qu.: 45.17   1st Qu.: 2.100  
 Median :0.5380   Median :6.208   Median : 76.80   Median : 3.207  
 Mean   :0.5547   Mean   :6.285   Mean   : 68.52   Mean   : 3.795  
 3rd Qu.:0.6240   3rd Qu.:6.623   3rd Qu.: 93.97   3rd Qu.: 5.188  
 Max.   :0.8710   Max.   :8.780   Max.   :100.00   Max.   :12.127  
                                  NA's   :20                       
     RAD              TAX           PTRATIO             B        
 Min.   : 1.000   Min.   :187.0   Min.   :12.60   Min.   :  0.32  
 1st Qu.: 4.000   1st Qu.:279.0   1st Qu.:17.40   1st Qu.:375.38  
 Median : 5.000   Median :330.0   Median :19.05   Median :391.44  
 Mean   : 9.549   Mean   :408.2   Mean   :18.46   Mean   :356.67  
 3rd Qu.:24.000   3rd Qu.:666.0   3rd Qu.:20.20   3rd Qu.:396.23  
 Max.   :24.000   Max.   :711.0   Max.   :22.00   Max.   :396.90  

    LSTAT             MEDV      
 Min.   : 1.730   Min.   : 5.00  
 1st Qu.: 7.125   1st Qu.:17.02  
 Median :11.430   Median :21.20  
 Mean   :12.715   Mean   :22.53  
 3rd Qu.:16.955   3rd Qu.:25.00  
 Max.   :37.970   Max.   :50.00  
```

```
 NA's   :20
```

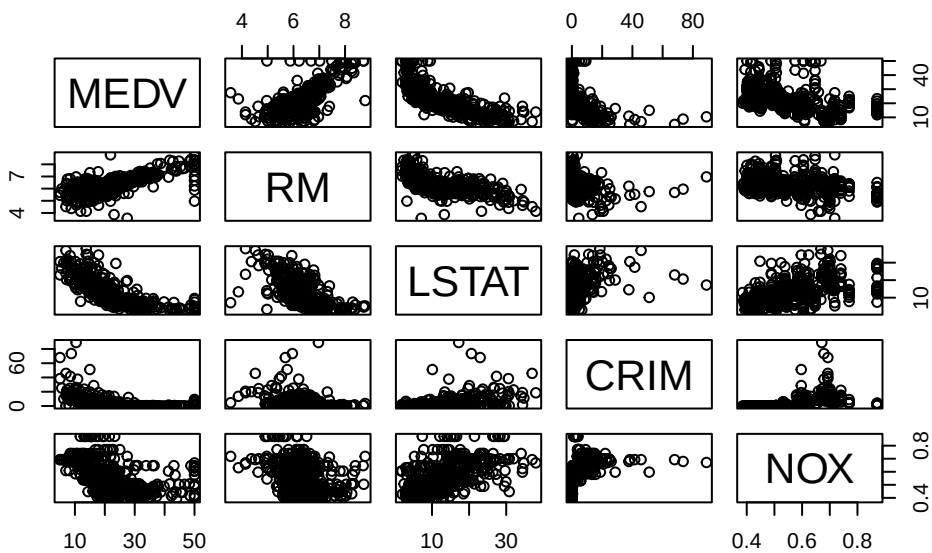
Correlation Matrix

```r
cor(data)
```

```
        CRIM ZN INDUS CHAS        NOX         RM AGE        DIS        RAD
CRIM       1 NA    NA   NA         NA         NA  NA         NA         NA
ZN        NA  1    NA   NA         NA         NA  NA         NA         NA
INDUS     NA NA     1   NA         NA         NA  NA         NA         NA
CHAS      NA NA    NA    1         NA         NA  NA         NA         NA
NOX       NA NA    NA   NA  1.0000000 -0.3021882  NA -0.7692301  0.6114406
RM        NA NA    NA   NA -0.3021882  1.0000000  NA  0.2052462 -0.2098467
AGE       NA NA    NA   NA         NA         NA   1         NA         NA
DIS       NA NA    NA   NA -0.7692301  0.2052462  NA  1.0000000 -0.4945879
RAD       NA NA    NA   NA  0.6114406 -0.2098467  NA -0.4945879  1.0000000
TAX       NA NA    NA   NA  0.6680232 -0.2920478  NA -0.5344316  0.9102282
PTRATIO   NA NA    NA   NA  0.1889327 -0.3555015  NA -0.2324705  0.4647412
B         NA NA    NA   NA -0.3800506  0.1280686  NA  0.2915117 -0.4444128
LSTAT     NA NA    NA   NA         NA         NA  NA         NA         NA
MEDV      NA NA    NA   NA -0.4273208  0.6953599  NA  0.2499287 -0.3816262
               TAX    PTRATIO          B LSTAT       MEDV
CRIM            NA         NA         NA    NA         NA
ZN              NA         NA         NA    NA         NA
INDUS           NA         NA         NA    NA         NA
CHAS            NA         NA         NA    NA         NA
NOX      0.6680232  0.1889327 -0.3800506    NA -0.4273208
RM      -0.2920478 -0.3555015  0.1280686    NA  0.6953599
AGE             NA         NA         NA    NA         NA
DIS     -0.5344316 -0.2324705  0.2915117    NA  0.2499287
RAD      0.9102282  0.4647412 -0.4444128    NA -0.3816262
TAX      1.0000000  0.4608530 -0.4418080    NA -0.4685359
PTRATIO  0.4608530  1.0000000 -0.1773833    NA -0.5077867
B       -0.4418080 -0.1773833  1.0000000    NA  0.3334608
LSTAT           NA         NA         NA     1         NA
MEDV    -0.4685359 -0.5077867  0.3334608    NA  1.0000000
```

Scatter Plot

```
pairs(data[,c("MEDV","RM","LSTAT","CRIM","NOX")])
```



Multiple linear Regression Model

- Regression model

$$MEDV = \beta_0 + \beta_1 RM + \beta_2 LSTAT + \beta_3 CRIM + \beta_4 NOX$$

```
model <- lm(MEDV ~ RM + LSTAT + CRIM + NOX, data=data)

summary(model)
```

```
Call:
lm(formula = MEDV ~ RM + LSTAT + CRIM + NOX, data = data)

Residuals:
```

```
    Min      1Q  Median      3Q     Max 
-17.077  -3.514  -1.183   1.851  29.110 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept) -1.01059    3.43761  -0.294  0.76891    
RM           5.03460    0.46017  10.941  < 2e-16 ***
LSTAT       -0.55863    0.05467 -10.217  < 2e-16 ***
CRIM        -0.10431    0.03301  -3.161  0.00168 ** 
NOX         -1.25103    2.74776  -0.455  0.64911    
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 5.49 on 461 degrees of freedom
  (40 observations deleted due to missingness)
Multiple R-squared:  0.6401,	Adjusted R-squared:  0.637 
F-statistic:   205 on 4 and 461 DF,  p-value: < 2.2e-16
```

Understanding the Hat Matrix

In linear regression:

$$\hat{y} = X(X^T X)^{-1} X^T y$$

The **Hat Matrix**:

$$H = X(X^T X)^{-1} X^T$$

Construct Design Matrix

```r
X <- model.matrix(model)

head(X)
```

```
  (Intercept)    RM LSTAT    CRIM   NOX
```

```
1           1 6.575  4.98 0.00632 0.538
2           1 6.421  9.14 0.02731 0.469
3           1 7.185  4.03 0.02729 0.469
4           1 6.998  2.94 0.03237 0.458
6           1 6.430  5.21 0.02985 0.458
7           1 6.012 12.43 0.08829 0.524
```

Compute Hat Matrix Manually

```r
H <- X %*% solve(t(X) %*% X) %*% t(X)

dim(H)
```

```
[1] 466 466
```

- Extract Levrage Values

```r
leverage <- diag(H)

head(leverage)
```

```
          1           2           3           4           6           7
0.005691995 0.003389691 0.006438337 0.006432836 0.005327368 0.003079703
```

Residual Analysis

```r
#residual s
res <- residuals(model)

print(res)
```

```
          1           2           3           4           6           7
 -4.63622290  -4.02110396   2.37783136   1.39715893   0.82465288   1.25112615
          8           9          10          11          12          13
  8.40548497   6.55824760  -0.09124410  -3.99200294  -2.26147278   2.50327784
         14          15          16          17          18          19
```

-3.18720383	-5.00916545	-2.99115817	-1.31097896	-2.69666227	1.02901538
20	21	22	23	24	25
-2.57242567	-0.88601992	-0.93297381	-3.45268956	-1.87374500	-3.35734384
26	27	28	29	30	31
-3.29436644	-2.63904183	-4.20770028	-5.38495438	-4.12029326	-1.63594516
32	33	34	35	37	38
-6.96056127	0.55230572	-3.54754721	-3.97643308	-1.38808538	-1.90999772
39	40	41	42	43	44
1.97563215	1.55899549	2.19256600	-3.19613791	-0.92699790	-0.82601328
45	46	47	48	49	50
-2.43620489	-2.01403054	0.36515934	-1.66136420	6.02724323	1.84000478
51	52	53	55	56	57
-1.23866230	-3.45436641	-3.26533778	-0.95102681	3.10730470	-2.68690474
58	59	60	61	62	63
1.01699696	-2.21199467	-3.50141014	-1.26473568	-4.37454706	-4.95498790
64	65	66	67	68	69
-2.14646513	3.26435573	-4.04660234	-2.50177020	-1.54016584	-1.91464969
70	71	72	73	75	76
-2.28261978	-2.81656187	-1.24814150	-3.11439035	-2.12900607	-3.68611873
77	78	79	81	82	83
-3.35754395	-2.80892169	-1.71892405	-1.36479025	-3.84252344	-1.62670885
84	85	86	87	89	90
-2.40582827	-1.31444401	-1.55326057	0.97860996	-6.97673755	-2.12787555
91	92	93	94	95	96
-3.15825719	-4.03939370	-3.37901255	-1.20674230	-3.34975599	0.34069264
97	98	99	100	101	102
-1.71405228	2.00752791	7.99955402	0.89577364	0.57118592	-1.68187605
103	104	105	106	107	108
-6.02336747	-2.40611820	-2.38475282	0.91820284	2.22115690	-0.90105050
109	110	111	112	113	114
-4.26510253	-1.58515565	-0.55473998	-3.62621945	-0.20638316	-0.70566925
115	118	119	120	121	122
-5.43893661	-3.64876454	1.13154273	-0.24029633	2.21908731	-0.21113575
123	124	125	126	127	128
2.25218223	3.76539174	0.77007132	1.29130712	4.44699477	-1.04080632
129	130	131	132	133	134
-3.95497355	-1.95154417	-4.44797326	-3.48437349	-1.01569937	-0.68954990
135	136	137	138	139	140
-1.82118919	-2.47035714	-1.24980410	-5.41487135	-2.46029451	-1.00742066
141	142	143	144	145	146
-1.76545363	10.31506882	3.62728902	5.35757036	5.86795318	0.81635954
147	148	149	150	151	152
-1.10844525	8.64334385	9.85447297	1.58946503	0.82794529	2.06799130

153	154	155	156	157	158
-0.94502946	1.80271338	-3.16281166	-4.51320315	-2.07060853	10.80397196
159	160	161	162	163	164
-0.74035083	-3.09807901	0.50651405	15.18239437	13.75134290	11.61577029
165	166	167	168	169	170
1.73130650	1.83656217	14.12472148	2.94871267	0.19460684	-1.58480923
171	172	173	174	175	176
-2.22378803	-1.77426293	4.91665718	-1.99377701	0.14491626	1.07656626
177	178	179	180	181	182
0.19541684	-2.02544296	0.88390699	6.50109885	6.55753706	12.17476433
183	185	186	187	188	189
6.20060970	7.62554846	7.59552490	14.68686992	3.15249472	0.91091626
190	191	194	195	196	198
3.30342628	6.42024777	1.18915365	-0.18795681	13.55159644	0.84978215
199	200	201	202	203	204
3.19638287	3.84904600	0.98062820	-1.23921795	7.25604871	12.62645087
205	206	207	209	210	211
12.69409115	0.65007739	0.32561724	3.69625984	7.66174554	2.98192144
212	213	214	215	216	217
7.15060278	3.76396600	2.88139031	14.61292755	0.80934426	2.90681649
218	219	220	221	222	223
2.37929786	3.26000634	-1.50929680	-1.18897820	4.34239244	0.12410542
224	225	226	228	231	232
2.73559360	7.17070633	10.35562685	0.77416867	2.39325002	-0.99425647
233	234	235	236	238	239
2.81119975	10.66565991	1.32584339	1.11667453	-1.20402704	-3.82172680
240	241	243	244	245	246
-4.28577960	-4.80855683	-1.98537030	-4.02757675	-2.00432867	2.16318171
247	248	249	250	251	252
0.25096124	-3.60501763	-1.00247767	-2.38815181	-3.39911777	-4.03517149
253	254	255	256	257	258
-1.89540645	4.78503465	-3.67511578	-2.01127166	9.71450883	10.92280814
259	260	261	262	264	265
5.32222777	1.39631799	4.76939687	11.17118290	2.30218519	6.62296596
266	267	268	269	270	271
2.54478523	5.55145750	14.17438788	9.44310116	0.12102875	0.50188137
273	274	275	276	277	279
-2.59497752	1.76888389	1.92380820	0.73736778	1.57398763	2.06039771
280	281	282	283	284	285
5.10026663	9.69825964	4.45154236	10.76300032	13.38992958	2.41195073
286	287	288	290	291	292
-4.39235712	-2.54815365	-2.54991444	-1.41782452	-2.65375461	4.83437302
293	294	295	296	297	298

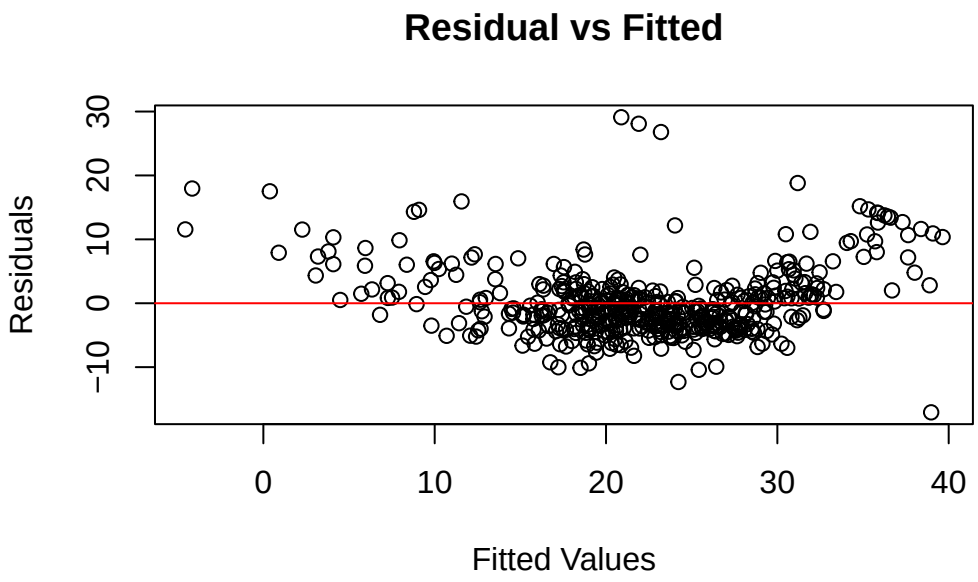
-1.32531316	-0.58802596	-1.17729584	0.05232847	-0.18042256	1.57041544
299	300	301	302	304	305
-5.15040671	-2.28392169	-4.88624776	-4.31503992	2.22607073	5.14776716
306	307	308	309	310	311
1.68645443	1.26650158	-0.46926800	-6.32582664	-2.46947343	0.09012348
312	313	314	315	316	317
-3.60761381	-2.65777655	-3.81436882	-2.34845593	-4.38052001	-0.01052552
318	319	320	321	322	323
1.31128461	-1.51030471	-0.92436931	-2.88536975	-3.51652968	-4.04858355
324	325	326	327	328	329
-2.02218702	-2.21521730	-3.29231819	-3.68381828	0.37192885	-3.08050615
330	331	332	333	334	335
-3.59069564	-4.46381825	-3.11996877	-5.02756024	-4.76534648	-5.63880249
336	337	338	339	340	341
-3.16024274	-2.91502963	-3.62176945	-3.49236983	-4.03040257	-4.49551780
342	343	344	345	346	347
0.87623307	-9.93289559	-4.18197541	0.78670097	-5.32889661	-3.84622312
348	349	352	353	354	355
-4.60333594	-4.00202757	-4.42270467	-5.13953047	0.26632932	-4.28220565
356	357	358	359	360	361
-4.63543509	-0.73205477	-0.70775062	0.78260751	1.32479848	-0.41196530
362	363	364	365	366	367
-1.27016487	1.85451258	-1.82305773	-17.07740517	15.93332451	7.02861922
368	369	371	372	373	374
14.30847616	29.11010796	18.81284842	26.79173930	28.09104616	11.52897899
375	376	378	379	380	381
17.95408243	-10.41446668	-6.16493604	-1.46983524	-5.24948979	-3.93547439
382	383	385	386	387	388
-6.76950292	-0.54313801	7.90129177	1.48395830	7.30815299	4.34122419
389	390	391	392	393	396
6.11805378	-1.25232662	-1.49189024	5.65404956	1.78552869	-7.12826113
397	398	399	400	401	402
-6.43579227	-6.62801002	0.51314702	-3.50822510	-5.09729591	-10.02044079
403	404	405	406	407	409
-5.91720879	-3.12115720	2.15867882	-1.81154619	8.10136363	6.19837422
411	412	413	414	415	416
-1.24475789	-1.23798454	17.51984672	6.31067281	11.55647845	-5.21777694
417	418	419	420	421	422
-9.24729174	3.14451154	-0.14137176	-10.11176143	-4.12074910	-4.62588137
423	424	425	426	428	429
3.27682840	-1.80140292	-4.07306401	-4.24442536	-6.42451658	-5.52894881
430	432	433	434	435	436
-6.33462903	-6.51109811	-7.11355930	-6.55689794	-7.72639125	-3.87441698

```
        437          438          439          440          443          444
-9.40457055  -3.97744059   0.88755720   0.16777652  -1.11501792  -3.74313737
        445          446          447          448          449          450
-2.11083403  -4.27287147  -4.49402539  -6.71569896  -4.03524627  -5.83217287
        451          452          453          454          455          456
-8.23262542  -5.93053435  -3.52172922  -7.30631543  -5.62577816  -6.22443051
        457          458          459          460          462          463
-4.37754820  -4.16412533  -5.04515631   0.20853584  -3.92882028  -1.89151770
        464          465          467          468          469          470
-4.33222389   0.17306111   0.83841994   2.98986246   2.75394175   2.68286375
        471          472          473          475          476          477
 0.14158331  -2.47352035   0.92269773  -0.80763627  -1.85204719  -3.22219694
        478          479          480          481          482          483
 2.55793294  -3.62067083   0.63708543  -0.14259853  -3.68816533  -4.35933377
        484          485          486          487          488          489
 0.58171753   0.48208364  -2.54491169  -0.97955102  -0.48860162  -0.38158507
        490          491          492          493          494          495
-5.07527708   0.83314952  -4.64401130  -0.78018618   1.53723562   4.02838487
        496          497          498          499          500          501
 6.14687786   6.14567313  -1.22331076  -0.11833831  -0.34133544  -3.77247747
        503          504          505          506
-3.40722533  -6.33693537  -6.84629027 -12.32424961
```

Plot Residuals vs Fitted

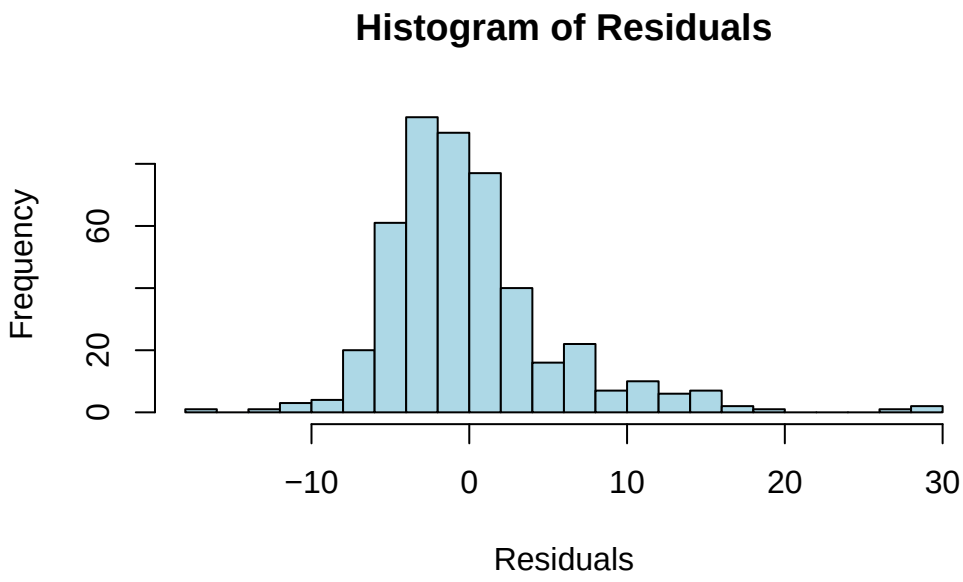
```r
plot(model$fitted.values,
     res,
     xlab="Fitted Values",
     ylab="Residuals",
     main="Residual vs Fitted")

abline(h=0,col="red")
```



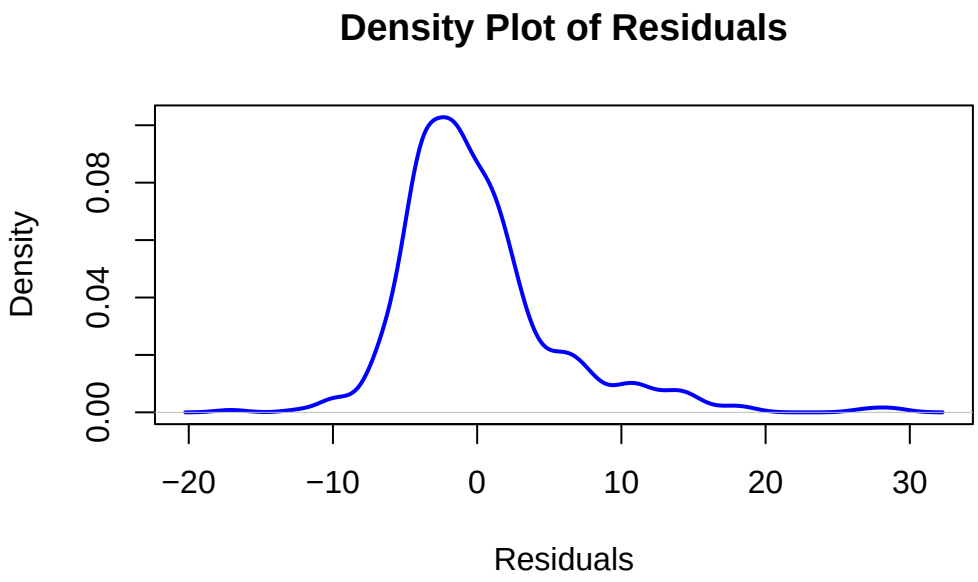
- **Histogram of Residuals**

```
hist(res,
     breaks = 20,
     col = "lightblue",
     main = "Histogram of Residuals",
     xlab = "Residuals")
```



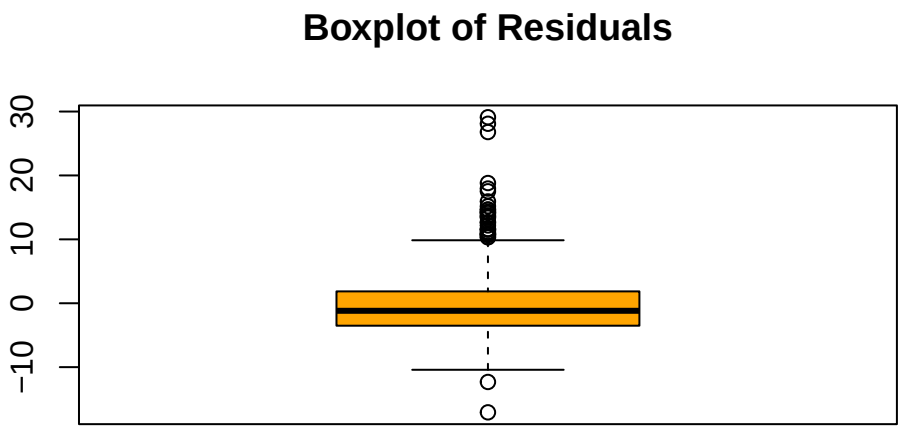
Density plot

```
plot(density(res),
     main = "Density Plot of Residuals",
     xlab = "Residuals",
     col = "blue",
     lwd = 2)
```



Boxplot of residual

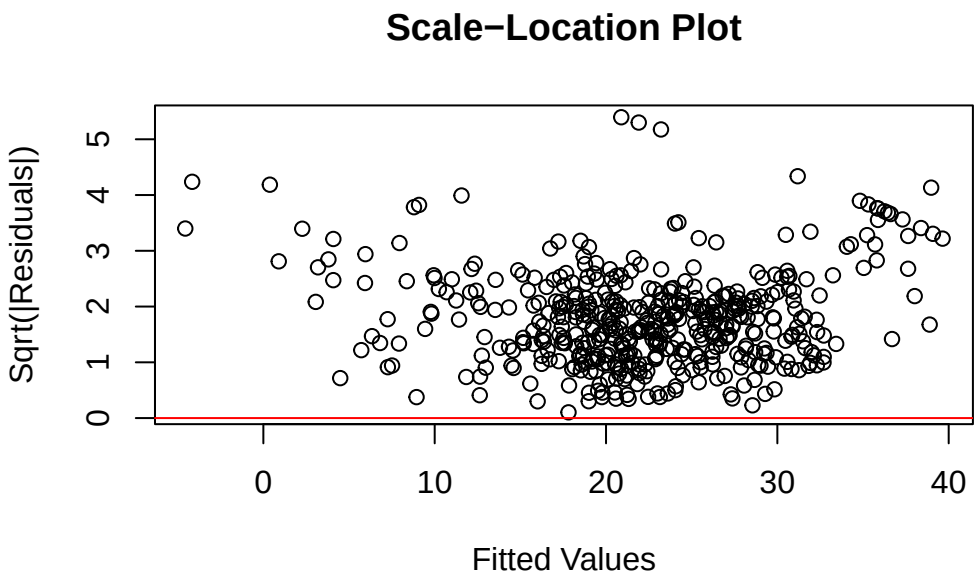
```
boxplot(res,
        main = "Boxplot of Residuals",
        col = "orange")
```



Scale-Location Plot

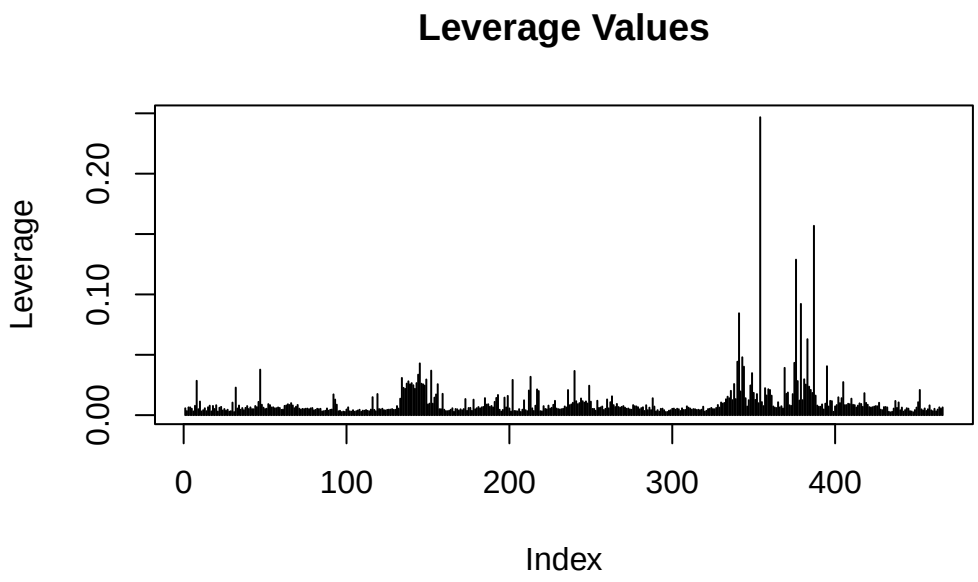
```
plot(model$fitted.values,
     sqrt(abs(res)),
     xlab = "Fitted Values",
     ylab = "Sqrt(|Residuals|)",
     main = "Scale-Location Plot")

abline(h=0,col="red")
```



Levrage Pot

```
plot(hatvalues(model),
     type="h",
     main="Leverage Values",
     ylab="Leverage")
```


Leverage Values
Leverage
0.00
0.10
0.20
0
100
200
300
400
Index